

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. Actual hard-sphere contacts are represented by a compatible Green trace and genuine open pre-contact evaluation maps. A surplus collision is imposed as a zero-set constraint; reflection and the later hard-sphere future are introduced only on that zero set. Genealogywise coarea bounds are summed by a fixed-horizon factorial majorant. The lower bound uses conservative smoothing and positive Maxwellian mixing to approximate every finite-action balanced pair inside the positive collision cone. This gives the grand-canonical and, after exact-number conditioning, microcanonical joint density–contact LDPs.

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CONTENTS

1. Round-seventeen positive closure: global pre-contact geometry and positive recovery	1
1.1. Compatible traces and exact balance	2
1.2. Open pre-contact charts	2
1.3. Fixed-horizon genealogy summation	3
1.4. Positive balanced approximation	4
1.5. Full grand-canonical and microcanonical LDPs	5
Scope and verification status	6

1. ROUND-SEVENTEEN POSITIVE CLOSURE: GLOBAL PRE-CONTACT GEOMETRY AND POSITIVE RECOVERY

The grand-canonical construction in this section is logically prior to B1. A surplus collision is imposed as the zero set of a genuine pre-contact map on an open trajectory-parameter domain. Reflection is introduced only on that

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zero set. The lower bound uses a density theorem for positive balanced pairs; no entropy penalty is asked to create a vector outside the positive collision cone.

1.1. Compatible traces and exact balance. Let $\mathfrak{X}_T$ be the space of finite space–time transport currents with velocity weight $1 + |v|^m$, $m > 8$, whose distributional transport divergence is a finite measure. At the hard-sphere boundary use the incoming normal-flux trace measure; the factor $((v - v_*) \cdot \omega)_+$ is part of that measure and is not multiplied a second time.

Theorem 1.1 (Closed Green trace graph). *The free-transport generator with specular binary boundary relation has a closed graph in $\mathfrak{X}_T$. For every smooth weighted test p , the empirical density π^ε and actual incoming contact measure Γ^ε satisfy exactly*

$$\langle p_T, \pi_T^\varepsilon \rangle - \langle p_0, \pi_0^\varepsilon \rangle - \int_0^T \langle (\partial_t + v \cdot \nabla_x) p, \pi_t^\varepsilon \rangle dt = \int \Delta_\omega p d\Gamma^\varepsilon.$$

The relation is closed under weighted weak convergence on entropy/moment sublevels.

Proof. Apply the divergence theorem on every free-flight cylinder. Interior time faces telescope. On a binary contact face the outgoing trace is the specular push-forward of the incoming flux trace, and their difference is $\Delta_\omega p$. Grazing and multiple contacts have zero Liouville flux. Closedness follows by velocity truncation, trace compactness for divergence-measure fields, and entropy uniform integrability of the collision weight; then let the truncation radius tend to infinity. $\square$

1.2. Open pre-contact charts. Fix a labelled collision genealogy G with strictly ordered creation contacts. Its open parameter space U_G consists of root positions and velocities, creation times and normals, and clustering parameters satisfying: strict time separation, non-grazing incidence, and absence of unintended contacts before the first surplus time. The collision-tree pseudo-trajectories are constructed recursively using only already imposed creation contacts. Before imposing the first surplus collision between labels a, b , define

$$\mathcal{F}_G(\theta, t, \omega) = x_a^\theta(t^-) - x_b^\theta(t^-) - \varepsilon\omega.$$

This map is smooth on the open set $U_G \times (t_-, t_+) \times \mathbb{S}^2$. The surplus-contact manifold is $\mathcal{F}_G^{-1}(0)$ with incoming sign. Only on this manifold are the specular reflection and the later chronology continued.

Lemma 1.2 (Global regular pre-contact atlas). *For every finite realized genealogy G , its regular surplus-contact set is covered by finitely many charts on which one 3×3 minor of $D_{(\theta, t, \omega)} \mathcal{F}_G$ is bounded below. The complement is a finite union of analytic sublevel strata. There are constants $C_G, \alpha_G > 0$ such that its δ -neighborhood has conditional trajectory measure at most $C_G \delta^{\alpha_G}$.*

Proof. The map is analytic between collision times. At a regular zero, time changes the component of relative position parallel to relative velocity, the two normal coordinates change the tangent components of $-\varepsilon\omega$, and a root or last-creation velocity changes the remaining transverse component through the true hard-sphere Jacobi map. Non-grazing incidence makes these three directions independent. For every realized combinatorial pattern, a witness is constructed inductively: place the creation collisions in disjoint spatial tubes with separated time intervals, solve the incoming linear flight equations backwards, and choose the final root velocity to hit the surplus contact. Strict inequalities exclude unintended contacts and persist on an open neighborhood. Thus the relevant analytic minor is not identically zero. Compact truncation and analytic sublevel estimates give C_G, α_G ; finitely many minors cover the regular zero set. $\square$

Lemma 1.3 (Physical first-surplus coarea gain). *For fixed G , after integrating the true incoming variables and imposing $\mathcal{F}_G = 0$ by coarea, the first surplus contact contributes at most*

$$C_G(1 + T + \mathbb{V})^{C_G} \varepsilon^{\alpha_G}$$

relative to the corresponding creation tree. The estimate includes bounded actual-contact marks and retains the true reflected future.

Proof. On the regular charts, coarea divides by a minor bounded away from zero and the contact tube has transverse volume $O(\varepsilon^2)$. On the analytic sublevel strata choose the threshold optimizing the regular tube bound and the sublevel measure from Lemma 1.2. Reflection is then applied on the zero set and is a smooth measure-preserving involution, so all later contacts are the physical future. Bounded marks change only the constant. $\square$

1.3. Fixed-horizon genealogy summation. Let $W_G^\varepsilon(h, \psi)$ be the absolute connected trajectory weight with a path source h and a mark on every actual incoming contact ψ . Creation-tree coarea and the Gaussian velocity activity give

$$\sum_{|G|=k} W_G^\varepsilon(h, \psi) \leq k! C^k T^{k-1}$$

on a fixed short-time source ball.

Theorem 1.4 (Two-limit all-contact majorant). *For every source radius R and fixed short horizon T , there are finite-genealogy approximations $Q_{T,K}^\varepsilon$ such that*

$$\lim_{K \rightarrow \infty} \limsup_{\varepsilon \downarrow 0} \frac{1}{\mu_\varepsilon} |\log Z_T^\varepsilon - \log Z_{T,K}^\varepsilon| = 0,$$

and, for fixed K , all surplus-contact contributions vanish as $\varepsilon \downarrow 0$. The statement remains true after any fixed number of source derivatives. No genealogy-uniform Łojasiewicz exponent is assumed.

Proof. First truncate the number of labels, contacts, and velocity. Only finitely many genealogies remain, so Lemma 1.3 may be applied with the minimum of their positive exponents. Then let $\varepsilon \rightarrow 0$. The discarded genealogy tail is bounded in normalized log partition function by the tail of the absolutely convergent series $\sum_k C^k T^{k-1}$. Letting $K \rightarrow \infty$ after the Boltzmann–Grad limit gives the assertion. Cauchy estimates on a slightly larger source ball give the derivative statement. Time slicing repeats the argument on finitely many blocks and permits every bounded real source on the fixed horizon. $\square$

Theorem 1.5 (Grand-canonical actual-contact pressure). *The normalized grand-canonical log-Laplace functional of $(\pi^\varepsilon, \Gamma^\varepsilon)$ converges locally uniformly, with all fixed source derivatives, to the unique analytic mild solution generated by*

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \frac{1}{2} \int f f_* B(e^{\Delta p + \psi} - 1).$$

Every ψ -derivative marks an actual contact. This theorem uses no micro-canonical input.

Proof. The logarithm is the sum of connected real-trajectory genealogies. Apply Theorem 1.4; creation trees converge by ordinary collision coarea and the first-surplus estimate removes cyclic corrections. Selecting the last creation contact factorizes the two rooted subtrees and gives the displayed Hamiltonian. Normal convergence proves analyticity and uniqueness by Picard iteration in the trajectory norm. $\square$

1.4. Positive balanced approximation. For a density f , let

$$dA_f = \frac{1}{2} f f_* B dt dx dv dv_* d\omega.$$

A finite-action pair is balanced and has $\Gamma = qA_f$ with $q \geq 0$ and $\int \ell(q) dA_f < \infty$.

Let S_δ be the conservative smoothing operator obtained by convolving in (t, x) and, at fixed center-of-mass and relative speed, convolving the relative direction on the collision sphere. It commutes with the elastic pre/post involution and with Δ^* .

Theorem 1.6 (Rate-dense strictly positive balanced pairs). *For every finite-action balanced pair (f, Γ) , there are smooth pairs (f_n, Γ_n) such that:*

- (i) $f_n > 0$, $\Gamma_n = q_n A_{f_n}$, and $0 < c_n \leq q_n \leq C_n < \infty$;
- (ii) the exact balance equation and all collision symmetries hold;
- (iii) $(f_n, \Gamma_n) \rightarrow (f, \Gamma)$ in the weighted weak path topology;
- (iv)

$$I_0(f_{n,0}) + \int \ell(q_n) dA_{f_n} \longrightarrow I_0(f_0) + \int \ell(q) dA_f.$$

Proof. First truncate velocities and replace the tail by a Maxwellian with the same mass, momentum, and energy. Mix the balanced pair with δ times a

strictly positive Maxwellian Boltzmann solution and its natural collision flow. Balance is linear, the numerator collision measure stays positive, and the mixed density is strictly positive. The perspective entropy is jointly convex, so the action changes by $o(1)$.

Apply S_δ simultaneously to the density current and collision current. Commutation with transport and Δ^* preserves balance exactly. On the compact velocity truncation, strict positivity makes $q_n = d\Gamma_n/dA_{f_n}$ bounded above and below. Standard approximation of perspective integrals gives action convergence. A diagonal removal of the Maxwellian admixture, smoothing, and truncation proves the theorem. No signed correction of a positive collision current is used. $\square$

Theorem 1.7 (Positive source exposure). *Every approximant in Theorem 1.6 is the unique controlled Boltzmann solution for the bounded contact source $\psi_n = \log q_n$ and its prepared initial activity. Under the corresponding microscopic tilt, $(\pi^\varepsilon, \Gamma^\varepsilon)$ concentrates at (f_n, Γ_n) , and the logarithmic likelihood cost is exactly*

$$I_0(f_{n,0}) + \int \ell(q_n) dA_{f_n}.$$

Proof. The first derivative of the analytic pressure in Theorem 1.5 solves the controlled Boltzmann equation with intensity $e^{\psi_n} A_f = q_n A_f$. The prescribed pair satisfies that equation and the same initial condition, so short-time uniqueness identifies it. The second derivative and exponential moment bounds give concentration. The Radon–Nikodym formula for the contact tilt is $\int \log q_n d\Gamma_n - \int (q_n - 1) dA_{f_n}$, which is the perspective entropy. $\square$

1.5. Full grand-canonical and microcanonical LDPs.

Theorem 1.8 (Grand-canonical joint LDP). *At speed μ_ε , $(\pi^\varepsilon, \Gamma^\varepsilon)$ satisfies a full good LDP with rate*

$$I_T^{\text{gc}}(f, \Gamma) = I_0^{\text{gc}}(f_0) + \begin{cases} \int \ell(d\Gamma/dA_f) dA_f, & (f, \Gamma) \text{ balanced and } \Gamma \ll A_f, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finite action excludes grazing, multiple-contact, and corner mass.

Proof. Finite collections of density/contact tests have log-Laplace limits by Theorem 1.5. Their convex dual imposes the exact balance and gives the pointwise entropy conjugate. Exponential velocity, contact-mass, and time-modulus bounds give projective exponential tightness; Dawson–Gartner gives the upper bound and a preliminary projective rate. The regular tilted lower bound is Theorem 1.7; Theorem 1.6 extends it to every finite-rate pair. Thus the projective rate equals the displayed action, whose sublevels are compact. If a limit charged a contact stratum on which $A_f = 0$, absolute continuity would fail and the entropy would be infinite. $\square$

Theorem 1.9 (Source-conditioned microcanonical LDP). *After the exact-number, momentum, energy, and macro shell theorem of B1 is proved from*

the grand-canonical pressure above, the primitive microcanonical law satisfies the same joint LDP with I_0^{gc} replaced by the B1 constrained initial rate. The multiplier is reoptimized for every density and contact source.

Proof. B1 supplies a locally uniform source-dependent shell coefficient with subexponential probability under the reoptimized tilted law. Divide the regular microscopic changes of measure in Theorem 1.7 by that coefficient. The upper bound and exponential tightness transfer by the same conditioning identity. The positive approximation theorem is unaffected by conditioning at time zero. $\square$

Theorem 1.10 (Round-seventeen B2 closure). *Actual hard-sphere contacts have a global pre-contact coarea construction, a fixed-horizon genealogy limit, an analytic grand-canonical marked pressure, a positivity-preserving rate-dense recovery theorem, and full grand-canonical and source-conditioned microcanonical joint LDPs.*

Proof. Combine Theorems 1.1, 1.4, 1.5, 1.6, 1.7, 1.8, and 1.9. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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